Trainings

Preparatory Steps

⚠ WARNING ⚠

Batteries can emit explosive gases. To reduce the possibility of personal injury, always ventilate the compartment before servicing the batteries. To reduce the possibility of arcing, remove the negative (-) battery cable first and attach the negative (-) battery cable last.

⚠ WARNING ⚠

Some state and federal agencies have determined that used engine oil can be carcinogenic and cause reproductive toxicity. Avoid inhalation of vapors, ingestion, and prolonged contact with used engine oil. If not reused, dispose of in accordance with local environmental regulations.

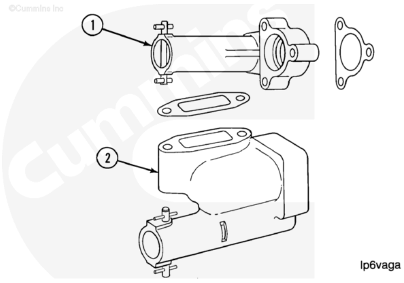
⚠ WARNING ⚠

To reduce the possibility of personal injury, avoid direct contact of hot oil with your skin.

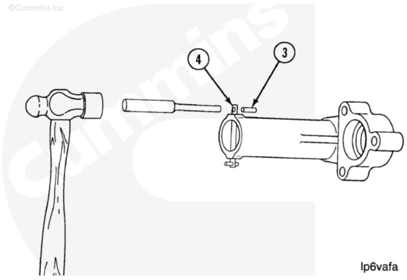
- Disconnect the batteries. Refer to the OEM service manual.
- Drain the oil from the engine. Refer to Procedure 007-037 in Section 7. (/qs3/pubsys2/xml/en/procedures/28/28-007-037-tr.html)
- Remove the oil pan. Refer to Procedure 007-025 in Section 7. (/qs3/pubsys2/xml/en/procedures/28/28-007-025-tr.html)
- Remove the oil transfer tubes. Refer to Procedure 007-040 in Section 7. (/qs3/pubsys2/xml/en/procedures/28/28-007-040-tr.html)

Disassemble

This procedure describes the high-pressure relief valve for the K38 (1), QSK38 (2), K50 (2), and QSK50 (2), engines. The function and procedure are the **same**. The K38 valve (1) is used as the example.

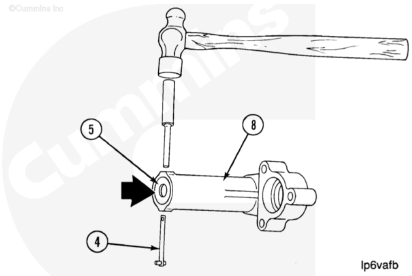


Use a small hammer and a drift. Remove **one** of the roll pins (3) from the retaining pin (4).



⚠ WARNING ⚠

The parts are under spring pressure. The parts can move suddenly with enough force to cause personal injury. Use a tool that will control the parts firmly.

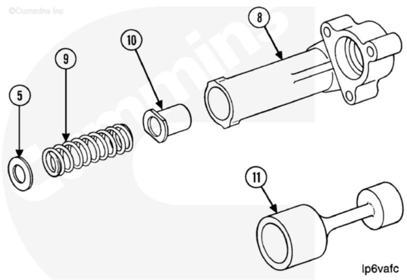


Push on the washer (5) until it does **not** touch the retaining pin.

Use a hammer and a drift. Remove the retaining pin (4) from the valve body (8).

Remove the hardened washer (5), the spring (9), and the plunger (10) from the body (8).

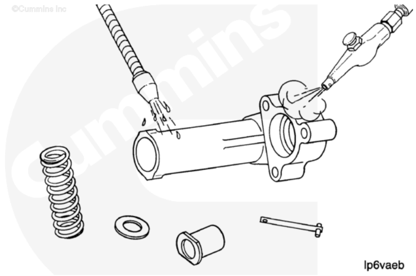
The plunger (11) for QSK38, K50, and QSK50 valves is different than the plunger for the K38 valves.



Clean

⚠ WARNING ⚠

When using solvents, acids or alkaline materials for cleaning, follow the manufacturers recommendations for use. Wear goggles and protective clothing to reduce the possibility of personal injury.



lp6vaeb

⚠ WARNING ⚠

Wear appropriate eye and face protection when using compressed air. Flying debris and dirt can cause personal injury.

Use solvent to clean the parts.

Dry with compressed air

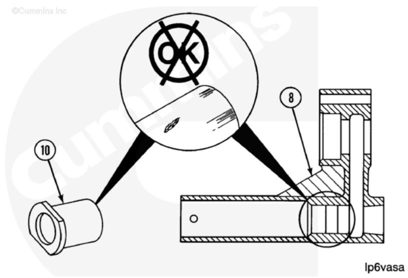
Check the parts for damage.

If the washer is worn, it **must** be replaced with the correct hardened washer.

Inspect

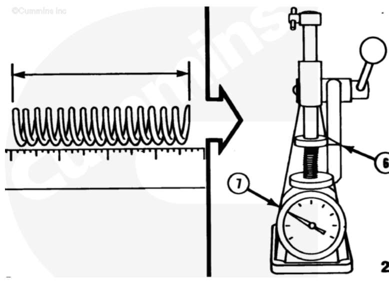
Check the bore in the plunger body and the plunger for damage.

If a fine grit crocus cloth does **not** remove the damage, the part **must** be replaced.



lp6vasa

Measure the free length of the spring. Use a spring tester to measure the spring force (7) at the working height (6).



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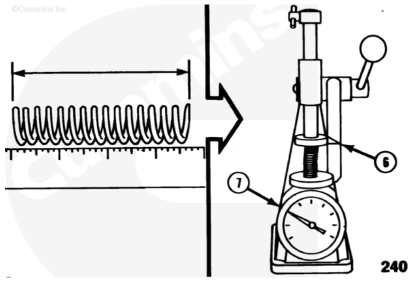
K38 High-Pressure Relief Valve Spring			
	mm		in
Free Length:	178.8	MIN	7.04
	182.6	MAX	7.19
Working Height (6):	117.09	NOM	4.61

K38 High-Pressure Relief Valve Spring

	n		lbf
Load (7):	729	MIN	164
	164	MAX	192

QSK38 and K50 High-Pressure Relief Valve Spring			
	mm		in
Free Length:	92.2	MIN	3.63
	94.7	MAX	3.73
Working Height (6):	49.3	NOM	1.94



	n		lbf
Load (7):	391	MIN	88
	463	MAX	104

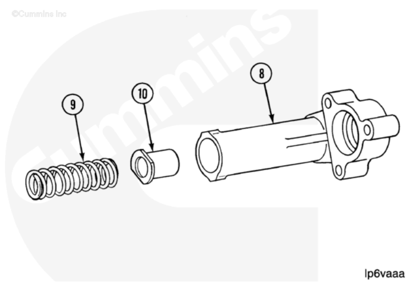
QSK50 High-Pressure Relief Valve Spring			
	mm		in
Free Length:	83.1	MIN	3.27
	85.6	MAX	3.37
Working Height (6):	49.3	NOM	1.94
Load (7):	445	MIN	100
	512	MAX	115

If the part is **not** within specifications, it **must** be replaced.

Assemble

Use clean engine oil to lubricate the plunger and the bore in the body.

Install the plunger (10) and the spring (9) in the body (8).



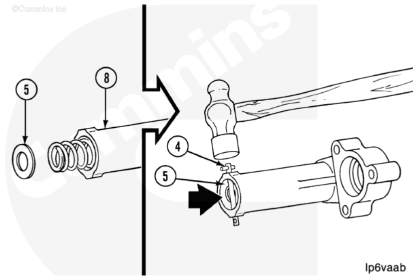
⚠ WARNING ⚠

The parts are under spring pressure. The parts can move suddenly with enough force to cause personal injury. Use a tool that will control the parts firmly.

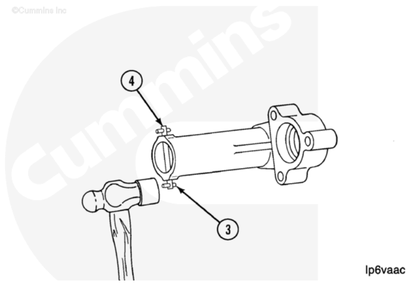
Place the hardened washer (5) on the end of the spring.

Push the parts into the bore until the washer is beyond the hole for the retaining pin.

Install the retaining pin (4) in the body. Use a small hammer and a drift to move the pin through the hole in the body until the hole in the pin for the roll pin is exposed.



Use a small hammer. Install the roll pin (3) in the hole in the retaining pin (4).



Finishing Steps

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- Install the oil transfer tubes. Refer to Procedure 007-040 in Section 7. (/qs3/pubsys2/xml/en/procedures/28/28-007-040-tr.html)
- Install the oil pan. Refer to Procedure 007-025 in Section 7. (/qs3/pubsys2/xml/en/procedures/28/28-007-025-tr.html)
- Fill the engine with clean lubricating oil. Refer to Procedure 007-037 in Section 7. (/qs3/pubsys2/xml/en/procedures/28/28-007-037-tr.html)
- Connect the batteries. Refer to the OEM service manual.
- Start the engine and check for leaks.

Last Modified: 04-Nov-2011